EV MARKET ANALYSIS DASHBOARD

FINAL PROJECT REPORT

Project: EV Market Analysis Dashboard

Technology: Tableau

Team ID: Individual project — no Team ID assigned.

Team Members: Individual project by Swati Jain.

Institution: Medi-Caps University, Indore — B.Tech, Computer Science and Engineering (CSE)

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TABLE OF CONTENTS

1. INTRODUCTION

1.1 Project Overview

1.2 Purpose

2. IDEATION PHASE

2.1 Problem Statement

2.2 Empathy Map Canvas

2.3 Brainstorming

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

3.2 Solution Requirement

3.3 Data Flow Diagram

3.4 Technology Stack

4. PROJECT DESIGN

4.1 Problem Solution Fit

4.2 Proposed Solution

4.3 Solution Architecture

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

7. RESULTS

7.1 Output Screenshots

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

8.2 Disadvantages / Limitations

9. CONCLUSION

9.1 Conclusion

10. FUTURE SCOPE

10.1 Future Enhancements

11. APPENDIX

11.1 Source Code (if any)

11.2 Dataset Link

11.3 GitHub & Project Demo Link

11.4 Project Details to Complete

1. INTRODUCTION

1.1 Project Overview

The EV Market Analysis Dashboard is a data-visualisation project developed in Tableau. It presents selected electric-vehicle market information through interactive and comparative visualisations. The dashboard is intended to make it easier to explore differences across vehicle categories, manufacturers, battery capacity, driving range, customer ratings, battery technology and manufacturing country.

The project uses a provided EV dataset as its data source. Tableau worksheets and dashboards are used to organise the information into visual summaries. The project work described in this report is based on the nine visualisation sheets prepared during development. Exact dataset counts and final dashboard evidence must be added from the completed Tableau workbook.

1.2 Purpose

The purpose of the project is to convert EV market data into understandable visual information that can support exploration and comparison. The dashboard is designed to help a prospective EV buyer compare selected vehicle characteristics and help a market analyst explore manufacturer and category-level patterns. It is an analytical aid and does not independently provide purchase recommendations or prove market-wide conclusions beyond the supplied dataset.

2. IDEATION PHASE

2.1 Problem Statement

Electric-vehicle information can be difficult to compare when specifications and market indicators are viewed separately. A user may need to examine vehicle categories, battery capacity, driving range and customer ratings together before making a comparison. Similarly, an analyst may need a consolidated view of manufacturer sales performance and country-level patterns.

Problem statement: How can the supplied EV dataset be presented in a clear Tableau dashboard so that users can explore vehicle characteristics and selected market-performance indicators in one place?

2.2 Empathy Map Canvas

The project’s draft empathy-map persona is a prospective EV buyer.

Says: “I want to compare EV options using clear information.”

Thinks: “Which vehicle category and manufacturer appear suitable based on the available specifications and ratings?”

Does: Compares categories, driving range, battery capacity, ratings and manufacturer-level visualisations.

Feels: May feel uncertain when information is scattered, and more confident when comparisons are presented clearly.

This persona is a design aid created for the project; it is not the result of a formal customer interview.

2.3 Brainstorming

The project ideation focused on combining related EV indicators in a single Tableau-based analytical view. The selected visualisation topics were:

- Sales by Category
- Battery Capacity by Category
- Customer Ratings by Category
- Driving Range by Category
- Top Manufacturers by Sales
- Customer Satisfaction by Manufacturer

• Battery Technology Distribution
• Manufacturer Sales Performance
• Sales by Manufacturing Country

These topics were selected to cover vehicle characteristics and selected market-performance comparisons. The final report should include a screenshot of the completed dashboard in this section or in Section 7.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The draft customer journey is organised into three stages:

1. Awareness and exploration: The user opens the dashboard and identifies the available EV categories and market indicators.

2. Evaluation and comparison: The user compares battery capacity, driving range, customer ratings, battery technology and manufacturer-level information.

3. Decision support: The user uses the visible comparisons as supporting information while considering EV options.

The dashboard informs exploration; the final purchase decision also depends on factors outside the scope of this dataset, such as budget, charging access, location and individual needs.

3.2 Solution Requirement

Functional requirements:
• Connect the supplied EV dataset to Tableau.
• Present the selected EV indicators in visual form.
• Provide a consolidated dashboard containing the project visualisations.
• Make labels and chart titles understandable.
• Provide interactive filters where they are present in the completed workbook.

Non-functional requirements:
• Keep the dashboard readable and consistently formatted.
• Use appropriate chart types for the data being compared.
• Maintain clear titles, legends and labels.
• Ensure the workbook opens and the visualisations render correctly.
• Avoid unsupported conclusions and clearly identify the dataset scope.

The exact filters and calculated fields must be confirmed from the final Tableau workbook before submission.

3.3 Data Flow Diagram

High-level data flow:

Provided EV dataset → Tableau data connection → data inspection/preparation → Tableau worksheets and visualisations → dashboard assembly → Tableau workbook (.twb) → Tableau Public publication (if published).

The data flow describes the Tableau workflow used for this project. It does not imply that a separate Snowflake database, SQL pipeline or automated ETL system was implemented.

3.4 Technology Stack

Primary tool: Tableau Desktop / Tableau Public for connecting to the dataset, creating worksheets and assembling the dashboard.

Input: The EV dataset supplied for the capstone project. Add the exact filename, format and source URL in the Appendix.

Output: Tableau workbook (.twb), dashboard screenshots and, if completed, a Tableau Public dashboard link.

Documentation: Word/PDF templates and project report.

Version numbers, operating system details and any additional tools should be added only if they were actually used.

4. PROJECT DESIGN

4.1 Problem Solution Fit

A Tableau dashboard is an appropriate solution for this project because the key task is to explore and compare several indicators in a supplied dataset. Visualisations allow users to identify differences and patterns more quickly than reading separate raw data tables. The solution is scoped to descriptive analysis of the available dataset.

4.2 Proposed Solution

The proposed solution is a Tableau dashboard built from nine worksheets. The worksheets cover category-level sales and specifications, customer ratings, driving range, manufacturer comparisons, battery technology distribution and manufacturing-country sales. The workbook is intended to provide a consolidated view of these topics.

The report should be updated with the exact final dashboard title, filter list, calculated fields (if any), and Tableau Public URL after checking the saved workbook.

4.3 Solution Architecture

The project architecture is a straightforward desktop visual-analytics workflow:

1. Obtain the capstone EV dataset.

2. Connect the dataset in Tableau.

3. Inspect fields and prepare data as required.

4. Build individual worksheets for the selected business questions.

5. Combine the worksheets into the dashboard.

6. Save the Tableau workbook.

7. Publish to Tableau Public, if completed, and share the public link.

No cloud database or Snowflake component is claimed in this architecture.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was organised into phases: ideation, requirement analysis, project design, planning, development and testing/documentation. The planning draft contains four proposed sprints and 12 user stories, totalling 47 story points, with an illustrative average planned velocity of 11.75 story points per sprint.

These figures are planning estimates from the prepared planning document, not verified actual team velocity or measured delivery performance. The proposed schedule dates should be replaced with the actual dates followed by the team. Add team member names, roles and the actual Team ID before final submission.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The project testing record is prepared around the Tableau workflow rather than Snowflake-specific database testing.

Checks to complete and document:
- Confirm the dataset connects and renders in Tableau.
- Verify that each of the nine worksheets opens and displays the intended visualisation.
- Check chart titles, labels, legends and formatting.
- Test each dashboard filter that is actually present.
- Check that the final dashboard layout is readable.
- Reopen the saved .twb workbook and verify that the dashboard renders.
- If published, open the Tableau Public link and verify access.

Actual test outcomes, load times, data-row counts, null/duplicate counts and screenshots have not been supplied in the current materials. Record measured values only after performing the tests. If Tableau Performance Recording was not used, state that detailed performance timings were not measured.

7. RESULTS

7.1 Output Screenshots

Insert genuine screenshots from the final Tableau workbook:
1. Full dashboard view showing the dashboard title and all visible charts.
2. Sales by Category.
3. Battery Capacity by Category.
4. Customer Ratings by Category.
5. Driving Range by Category.
6. Top Manufacturers by Sales.
7. Customer Satisfaction by Manufacturer.
8. Battery Technology Distribution.
9. Manufacturer Sales Performance.
10. Sales by Manufacturing Country.

Use screenshots from the actual workbook. Do not use placeholder or illustrative images as evidence. Add captions and briefly explain the insight each chart displays, using the actual values visible in Tableau.

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

Combines multiple EV indicators in one dashboard.

Makes category and manufacturer comparisons easier to explore.

Uses visual summaries to make the dataset more accessible.

Provides a reusable Tableau workbook for further analysis.

Can support exploratory discussions by prospective buyers and analysts.

8.2 Disadvantages / Limitations

Findings are limited to the fields, records and coverage of the supplied dataset.

Missing, outdated or inconsistent source data may affect interpretation.

The dashboard does not automatically account for every purchase factor, such as local charging infrastructure, ownership cost or personal driving needs, unless those fields exist in the dataset.

The report currently lacks verified numerical test results and final output screenshots.

Any interactivity depends on the filters and actions actually configured in the workbook.

9. CONCLUSION

9.1 Conclusion

The EV Market Analysis Dashboard demonstrates how Tableau can be used to organise a supplied electric-vehicle dataset into a set of comparative visualisations. The nine prepared sheets cover vehicle categories, battery capacity, customer ratings, driving range, manufacturers, battery technology and manufacturing country. The resulting dashboard is intended to make these indicators easier to explore together.

Before final submission, the report must be checked against the final workbook and completed with genuine screenshots, verified dataset details, actual Team ID and working publication/repository links.

10. FUTURE SCOPE

10.1 Future Enhancements

Add more recent and geographically diverse EV data.

Include price, charging time, charging infrastructure and ownership-cost indicators if reliable data becomes available.

Add further dashboard filters and drill-down interactions where useful.

Compare trends across multiple years if the dataset supports time-based analysis.

Improve accessibility through consistent labels, colour choices and explanatory tooltips.

Publish the dashboard and maintain a documented, versioned project repository.

11. APPENDIX

11.1 Source Code (if any)

The current project description is Tableau-focused. Attach the Tableau workbook (.twb) and any supporting files actually used. If no separate source-code files were written, state: “No separate program source code; the primary implementation is the Tableau workbook.”

11.2 Dataset Link

Dataset name: Dataset: Electric Vehicle (EV) market dataset used for the Tableau analysis; exact source filename was not available in the project details.
Dataset source / URL: Dataset source URL: Not provided in the available project details.
File format: Dataset format: Not confirmed from the available project details.
Records and fields: Record and field counts: Not provided; no figures have been assumed.

11.3 GitHub & Project Demo Link

GitHub repository: GitHub repository: EV_Market_Analysis_Dashboard (add the repository URL if required).
Tableau Public dashboard:
https://public.tableau.com/app/profile/swati.jain8829/viz/Book1_17935305808630/EVMarketAnalysisDashboard?publish=yes

Ensure each link is accessible to the evaluator before submission.

11.4 Project Details to Complete

Team ID: Individual project — no Team ID assigned.
Name: Swati Jain; individual project
Project guide / faculty: [Enter if required]
Actual project dates: [Enter dates followed]
Final dashboard screenshot: Insert the genuine full-dashboard screenshot from Tableau.
Testing evidence: Insert genuine Tableau testing screenshots and actual measured results.

Verified project dashboard link:
https://public.tableau.com/app/profile/swati.jain8829/viz/Book1_17935305808630/EVMarketAnalysisDashboard?publish=yes

Completion note: Team ID, dataset source/file details, verified data counts, final demonstration-video URL, and genuine dashboard/testing screenshots still need to be supplied before final submission. No unsupported metrics or test results have been invented.

Individual project details: Swati Jain | Enrollment No.: EN25CS3010952 | Team ID: Not applicable (individual project).

Dataset note: The report identifies the source as the EV dataset used for the Tableau analysis. The exact source URL, original filename, format, and row/field counts were not present in the supplied project information, so they are not guessed.

Demo video: Omitted as requested.